Aliasing Artifacts: Sampling and Quantization

Question: An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

Concept

Aliasing occurs when an image containing high-frequency details is sampled below the Nyquist rate. During downsampling, fine patterns are misrepresented as false patterns (jagged edges or moiré effects). Sampling determines how frequently pixels are recorded. If the sampling frequency is too low, high-frequency information folds into lower frequencies, producing aliasing. Quantization converts sampled intensity values into discrete levels. While low quantization can introduce banding, it is not the primary cause of aliasing in this example.

Corrective Approach

Apply a low-pass (anti-aliasing) filter such as Gaussian blur before downsampling. This removes high-frequency components that cannot be represented at the lower sampling rate. Then resize using an interpolation method such as INTER_AREA for better results.

Python Code

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

size = 200
high_freq_img = np.zeros((size, size), dtype=np.uint8)
for i in range(0, size, 4):
    high_freq_img[i:i+2, :] = 255
for j in range(0, size, 4):
    high_freq_img[:, j:j+2] = 255

alias_img = cv2.resize(high_freq_img, (size // 4, size // 4),
                       interpolation=cv2.INTER_NEAREST)

blurred_img = cv2.GaussianBlur(high_freq_img, (5, 5), 0)

anti_alias_img = cv2.resize(blurred_img, (size // 4, size // 4),
                            interpolation=cv2.INTER_AREA)

plt.figure(figsize=(15,5))
plt.subplot(131), plt.imshow(high_freq_img, cmap='gray')
plt.title('Original High-Frequency Image')
plt.axis('off')

plt.subplot(132), plt.imshow(alias_img, cmap='gray')
plt.title('Aliased Image (Undersampled)')
plt.axis('off')

plt.subplot(133), plt.imshow(anti_alias_img, cmap='gray')
plt.title('Anti-Aliased Image (Blurred then Undersampled)')
plt.axis('off')

plt.tight_layout()
plt.show()
```